

Introduction to Electrostatics

PHYS505 – Electromagnetic Theory

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Electric field

- Originally defined through force on unit charge q placed at point $\vec{x}$

$$\vec{F}(\vec{x}) = q \vec{E}(\vec{x})$$

- while assuming that q doesn't change the field distribution in its vicinity

$$E = \lim_{q \rightarrow 0} \frac{F}{q}$$

- can't really take that limit because charge comes in discrete units
- Coulomb's Law for force between charges q_1, q_2

$$\vec{F} = k q_1 q_2 \frac{\vec{x}_1 - \vec{x}_2}{|\vec{x}_1 - \vec{x}_2|^3}$$

- where k comes from units

$$k = \frac{1}{4\pi\epsilon_0} = \frac{\mu_0 c^2}{4\pi} = 10^{-7} c^2 = 89875517873681764 \simeq 9 \times 10^9$$

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Electric field

- also *unit* charge in SI is HUGE

$$E \simeq 9 \text{ GV/m}, \quad F \simeq 9 \text{ GN}$$

for $q_1, q_2 = 1 \text{ C}$, $r = 1 \text{ m}$

- air breaks down (⚡) at $\sim 3 \text{ MV/m}$, weight of $\sim \text{Mton}$
- Superposition for field due to point charges $\{q_i\}$ at $\{\vec{x}_i\}$

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n q_i \frac{\vec{x} - \vec{x}_i}{|\vec{x} - \vec{x}_i|^3}$$

- and for a continuous charge distribution $\rho(\vec{x})$

$$\sum dq_i \rightarrow \int \rho(\vec{x}') d^3x', \quad \vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} d^3x'$$

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Point charges

- Charge density $\rho(\vec{x})$ for a point charge q at $\vec{X}$ is a Dirac delta “function”

$$\delta(x - a) = 0, \quad x \neq a$$

$$\int_{x_1}^{x_2} \delta(x - a) dx = 1, \quad x_1 < a < x_2$$

$$\int f(x) \delta(x - a) dx = f(a) \text{ when } a \text{ in } (x_1, x_2)$$

$$\delta(\vec{x} - \vec{X}) = \delta(x_1 - X_1) \delta(x_2 - X_2) \delta(x_3 - X_3)$$

$$\delta(\vec{r} - \vec{R}) = \delta(x - X) \delta(y - Y) \delta(z - Z)$$

$$\rho(\vec{x}) = q \delta(\vec{x} - \vec{X})$$

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Consider the rectangular function of width ε

$$D_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon}, & |x| < \frac{\varepsilon}{2} \\ 0, & \text{elsewhere} \end{cases}$$

It is normalized

$$\int_{-\infty}^{\infty} D_\varepsilon(x) dx = 1$$

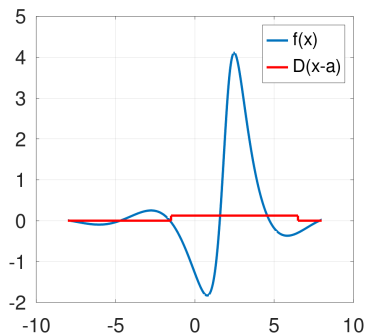
and can be used to compute the average value of an arbitrary function f over an interval of width ε around a point a as

$$\int_{-\infty}^{\infty} D_\varepsilon(x - a) f(x) dx = \frac{1}{\varepsilon} \int_{a-\varepsilon/2}^{a+\varepsilon/2} f(x) dx = \langle f \rangle$$

Sidebar – Dirac delta function

In the limit $\varepsilon \rightarrow 0$, D_ε behaves like the δ function.

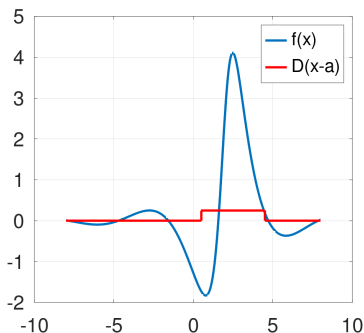
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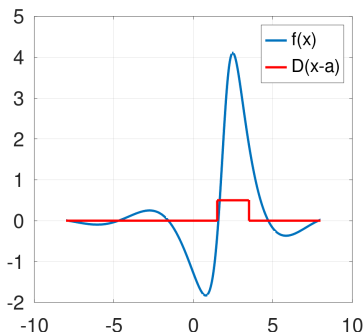
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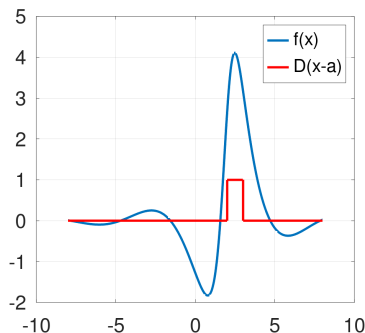
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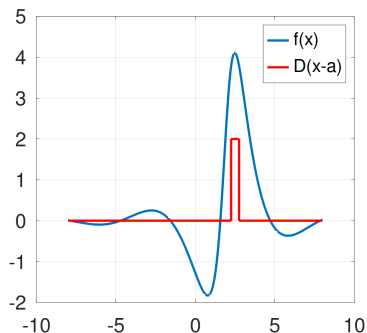
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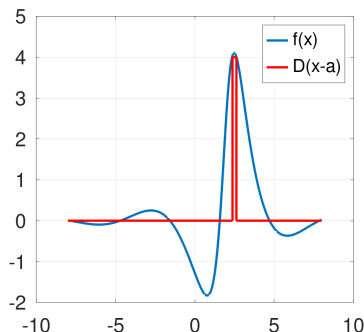
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Gauss' Law

- Consider a closed surface S with unit (outward) surface normal $\hat{\mathbf{n}}$ enclosing a volume V and a charge distribution ρ

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \rho(\vec{x}') \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3} d^3x'$$

- Gauss' Law follows from $1/r^2$ central force + superposition

$$\oint_S \vec{E} \cdot \hat{\mathbf{n}} \, da = \frac{1}{\epsilon_0} \int_V \rho(\vec{x}) \, d^3x = \frac{Q_{in}}{\epsilon_0} = \frac{1}{\epsilon_0} \sum_i q_i$$

- also applies to gravity

$$\oint \vec{g} \cdot \hat{\mathbf{n}} \, da = 4\pi G M \rightarrow g = G \frac{M_{\oplus}}{R_{\oplus}^2}$$

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Differential form of Gauss' Law

- Divergence theorem with closed surface S bounding volume V

$$\oint_S \vec{A} \cdot \hat{n} \, da = \int_V \vec{\nabla} \cdot \vec{A} \, d^3x$$

- applied to $\vec{E}$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

- or, in charge-free regions

$$\vec{\nabla} \cdot \vec{E} = 0$$

- also need curl to specify the vector field $\vec{E}$
- which follows from

$$\vec{\nabla} \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = - \frac{\vec{x} - \vec{x}'}{|\vec{x} - \vec{x}'|^3}$$
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Scalar potential

- giving

$$\vec{\nabla} \times \vec{E} = \vec{0}, \quad \vec{E} = -\vec{\nabla}\Phi$$

- with the scalar potential Φ defined through

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

- means potential difference is potential energy per unit charge

$$\begin{aligned} W_{\vec{A} \rightarrow \vec{B}} &= - \int_{\vec{A}}^{\vec{B}} \vec{F} \cdot d\vec{l} = -q \int_{\vec{A}}^{\vec{B}} \vec{E} \cdot d\vec{l} \\ &= q \int_{\vec{A}}^{\vec{B}} \vec{\nabla}\Phi \cdot d\vec{l} = q (\Phi_{\vec{A}} - \Phi_{\vec{B}}) \end{aligned}$$

- For a closed loop C that is the boundary of surface S

$$\oint_C \vec{E} \cdot d\vec{l} = 0 = \int_S (\vec{\nabla} \times \vec{E}) \cdot \hat{n} da \rightarrow \vec{\nabla} \times \vec{E} = \vec{0}$$

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- Fundamental theorem of calculus

$$\int_{\vec{A}}^{\vec{B}} \vec{\nabla} f \cdot d\vec{l} = \int_{\vec{A}}^{\vec{B}} df = f(\vec{B}) - f(\vec{A})$$

- means potential difference is potential energy per unit charge

$$\begin{aligned} W_{\vec{A} \rightarrow \vec{B}} &= - \int_{\vec{A}}^{\vec{B}} \vec{F} \cdot d\vec{l} = -q \int_{\vec{A}}^{\vec{B}} \vec{E} \cdot d\vec{l} \\ &= q \int_{\vec{A}}^{\vec{B}} \vec{\nabla}\Phi \cdot d\vec{l} = q (\Phi_{\vec{A}} - \Phi_{\vec{B}}) \end{aligned}$$

Scalar potential

- giving

$$\vec{\nabla} \times \vec{E} = \vec{0}, \quad \vec{E} = -\vec{\nabla}\Phi$$

- with the scalar potential Φ defined through

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

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- For a closed loop C that is the boundary of surface S

$$\oint_C \vec{E} \cdot d\vec{l} = 0 = \int_S (\vec{\nabla} \times \vec{E}) \cdot \hat{n} da \rightarrow \vec{\nabla} \times \vec{E} = \vec{0}$$

Surface distributions & boundary conditions

- A surface charge density $\sigma(\vec{x})$ results in a discontinuity in the electric field component normal to the surface with

$$\left[\vec{E}_{out}(\vec{x}) - \vec{E}_{in}(\vec{x}) \right] \cdot \hat{\mathbf{n}}(\vec{x}) = \frac{\sigma(\vec{x})}{\epsilon_0}$$

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- The potential is continuous (no singularity like for a point charge)
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Dipole layer

- Dipole layer on surface S with the dipole moment $D(\vec{x})$ as the limit of equal and opposite surface charge densities displaced by distance $d(\vec{x})$

$$D(\vec{x}) = \lim_{d(\vec{x}) \rightarrow 0} \sigma^{\pm}(\vec{x}) d(\vec{x})$$

- Potential due to dipole layer

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- Taylor expansion

$$\frac{1}{|\vec{x} + \vec{a}|} = \frac{1}{|\vec{x}|} + \vec{a} \cdot \vec{\nabla} \left(\frac{1}{|\vec{x}|} \right) + \dots$$

- gives, for small d ,

$$\frac{1}{|\vec{x} - \vec{x}' + d\hat{n}|} = \frac{1}{|\vec{x} - \vec{x}'|} + d\hat{n} \cdot \vec{\nabla} \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) + \mathcal{O}(d^2)$$

- and for the potential,

$$\begin{aligned}\Phi(\vec{x}) &= \frac{1}{4\pi\epsilon_0} \lim_{d \rightarrow 0} \int_S \sigma(\vec{x}') d(\vec{x}') \hat{n} \cdot \vec{\nabla} \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) da' \\ &= \frac{1}{4\pi\epsilon_0} \int_S D(\vec{x}') \hat{n} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) da'\end{aligned}$$

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- Potential for point dipole $\vec{p}$ at $\vec{x}'$

$$\Phi_p(\vec{x}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot (\vec{x} - \vec{x}')}{|\vec{x} - \vec{x}'|^3}$$

- Dipole layer potential is

$$\Phi(\vec{x}) = \int_S \Phi_p(\vec{x}), \text{ with } \vec{p}(\vec{x}') = D(\vec{x}') da' \hat{n}$$

- Also,

$$\begin{aligned} \hat{n} \cdot \vec{\nabla}' \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) &= \frac{\cos \theta da'}{|\vec{x} - \vec{x}'|^2} = -d\Omega \\ \rightarrow \Phi(\vec{x}) &= -\frac{1}{4\pi\epsilon_0} \int_S D(\vec{x}') d\Omega \end{aligned}$$

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- For constant surface dipole-moment density D , the potential is independent of the shape of S

$$\Phi(\vec{x}) = \frac{D}{4\pi\epsilon_0} \Delta\Omega_S$$

- Discontinuity for the potential at the dipole layer

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Poisson/Laplace equations

- Electric field determined by

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= \frac{\rho}{\epsilon_0} \\ \vec{\nabla} \times \vec{E} &= \vec{0}\end{aligned}$$

- with the scalar potential

$$\vec{E} = -\vec{\nabla}\Phi$$

- Combining these,

$$\vec{\nabla} \cdot (\vec{\nabla}\Phi) = -\nabla^2\Phi$$

- and

$$\nabla^2\Phi = -\frac{\rho}{\epsilon_0} \text{ (Poisson equation)}$$

- In charge-free regions

$$\nabla^2\Phi = 0 \text{ (Laplace equation)}$$

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Poisson/Laplace equations

- Using

$$\nabla^2 \left(\frac{1}{|\vec{x} - \vec{x}'|} \right) = -4\pi \delta(\vec{x} - \vec{x}')$$

- you can show that the expression for the potential

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

- satisfies the Poisson equation

$$\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$$

Exercise

Do it.

Green's theorem

- Why do we need differential equations (Poisson/Laplace) for the potential Φ if there's an integral formula that is straightforward to evaluate (numerically) given the (static) charge distribution ρ, σ ?

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{x}')}{|\vec{x} - \vec{x}'|} d^3x'$$

- Many practical problems involve boundary conditions
- with the charge distribution unknown
- except in a limited set of cases where it can be found using symmetry (eg. method of images)
- Example, with potential specified on conducting surfaces
- charges go wherever they need to on the surfaces of conductors to cancel the field inside

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Green's theorem

- Divergence theorem for vector field $\vec{A}$ in volume V bounded by closed surface S

$$\int_V \vec{\nabla} \cdot \vec{A} d^3x = \oint_S \vec{A} \cdot \hat{n} da$$

- For $\vec{A} = \phi \vec{\nabla} \psi$,

$$\vec{\nabla} \cdot \vec{A} = \vec{\nabla} \phi \cdot \vec{\nabla} \psi + \phi \nabla^2 \psi$$

$$\vec{A} \cdot \hat{n} = \phi \vec{\nabla} \psi \cdot \hat{n} = \phi \frac{\partial \psi}{\partial n} \text{ (normal derivative)}$$

- Plugging into div. thm., we get Green's 1st identity

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Green's theorem

- Interchanging ϕ and ψ and taking the difference ($\phi\psi$ minus SciFi), we get **Green's theorem** (aka Green's 2nd identity)

$$\int_V (\vec{\nabla} \phi \cdot \vec{\nabla} \psi + \phi \nabla^2 \psi) d^3x = \oint_S \phi \frac{\partial \psi}{\partial n} da$$

$$\int_V (\vec{\nabla} \psi \cdot \vec{\nabla} \phi + \psi \nabla^2 \phi) d^3x = \oint_S \psi \frac{\partial \phi}{\partial n} da$$

$$\rightarrow \int_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) d^3x = \oint_S \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) da$$

Green's theorem

- Using

$$\phi = \Phi, \quad \nabla^2 \phi = \frac{\rho}{\epsilon_0}, \quad \psi = \frac{1}{R} = \frac{1}{|\vec{x} - \vec{x}'|}, \quad \nabla^2 \psi = -4\pi \delta(\vec{x} - \vec{x}')$$

- we have

$$\begin{aligned} & \int_V \left[-4\pi \Phi(\vec{x}') \delta(\vec{x} - \vec{x}') + \frac{1}{R} \frac{\rho(\vec{x}')}{\epsilon_0} \right] d^3x' \\ &= \oint_S \left[\Phi(\vec{x}') \frac{1}{\partial n'} \left(\frac{1}{R} \right) - \frac{1}{R} \frac{\partial \Phi}{\partial n'} \right] da' \end{aligned}$$

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Green's theorem

- If $\vec{x}$ is in V ,

$$\Phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{x}')}{R} d^3x' + \frac{1}{4\pi} \oint_S \left[\frac{1}{R} \frac{\partial \Phi}{\partial n'} - \Phi(\vec{x}') \frac{\partial}{\partial n'} \left(\frac{1}{R} \right) \right] da'$$

- If $\rho = 0$ inside V , the potential anywhere within the volume V can be calculated using only the potential and its normal derivative on the surface S (Cauchy boundary conditions)
- ⚠ This is an overspecification – Φ and $\partial\Phi/\partial n$ are not independent
- If S is inflated toward $R \rightarrow \infty$, if the field on S falls off faster than $1/R$, the 1st term in the surface integral goes away ($da' \sim R^2 d\Omega$) and we are left with the usual integral formula for the potential (up to a constant if $\langle \Phi_S \rangle \neq 0$)

Green's theorem

- If $\vec{x}$ is in V ,

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Uniqueness of solutions

- The solution to the Poisson equation

$$\nabla^2\Phi = -\frac{\rho}{\epsilon_0}$$

within a volume is unique if we specify, on the bounding (closed) surface

- (only) the potential $\Phi \rightarrow$ Dirichlet boundary conditions
- (only) the electric field ($\partial\Phi/\partial n$ or, equivalently, σ)
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- Mixed boundary conditions, with Φ on part of S and $\partial\Phi/\partial n$ on the rest, also define a unique solution

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Uniqueness of solutions

- Assume Φ_1, Φ_2 are two different solutions satisfying the same boundary conditions
- Define $U = \Phi_2 - \Phi_1$, so $\nabla^2 U = 0$ inside V and either $U = 0$ (Dirichlet) or $\partial U / \partial n = 0$ (Neumann) on S
- Substitute $\phi = \psi = U$ into Green's 1st identity

$$\int_V (\vec{\nabla} U \cdot \vec{\nabla} U + U \nabla^2 U) d^3x = \oint_S U \frac{\partial U}{\partial n} da$$

- which, for either boundary condition, yields

$$\int_V |\vec{\nabla} U|^2 d^3x = 0 \rightarrow \vec{\nabla} U = \vec{0}$$

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Green('s) functions – the basic idea

- For a unit-charge point source at location $\vec{x}'$,

$$\rho_o(\vec{x}) = \delta(\vec{x} - \vec{x}')$$

- the Poisson equation $\nabla^2 \Phi = -\frac{\rho}{\epsilon_0}$ has the solution

$$\Phi_o = \frac{1}{4\pi\epsilon_0} \frac{1}{|\vec{x} - \vec{x}'|} \equiv G_o(\vec{x}, \vec{x}')$$

- The *free space Green function* $G_o(\vec{x}, \vec{x}')$ is the potential at $\vec{x}$ due to a point charge at $\vec{x}'$
- The solution Φ for an arbitrary charge distribution $\rho(\vec{x}')$ can be built up as

$$\Phi(\vec{x}) = \int_V \rho(\vec{x}') G_o(\vec{x}, \vec{x}') d^3x'$$

- We want to generalize this to problems with nontrivial bc by using superposition

$$G(\vec{x}, \vec{x}') = G_o(\vec{x}, \vec{x}') + F(\vec{x}, \vec{x}')$$

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- The $1/R$ potential for a point charge

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- Starting from the general expression for Φ with the bc included through the surface integral and using Green's theorem with $\phi = \Phi$, $\psi = G(\vec{x}, \vec{x}')$,

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- we can arrange the F part of G to cancel one of the terms in the $\oint_S$
- Using Dirichlet bc

$$G_D(\vec{x}, \vec{x}') = 0, \text{ for } \vec{x}' \in S$$

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- because it violates Gauss' Law

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- so we set it to a non-0 constant instead

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$$\Phi(\vec{x}) = \langle \Phi \rangle_S + \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}') G_N(\vec{x}, \vec{x}') d^3x' + \frac{1}{4\pi} \oint_S G_N \frac{\partial \Phi}{\partial n'} da'$$

Electrostatic potential energy

- Point charge q_i brought from infinity to point $\vec{x}_i$ in the presence of a local field distribution from a scalar potential Φ ($\Phi(|\vec{x}| \rightarrow \infty) \rightarrow 0$); work done on charge (and its potential energy)

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- If Φ is produced by a set of $(n - 1)$ point charges $\{q_j\}$ at positions $\vec{x}_j$

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- To build a system of n point charges, we can bring them from ∞ one-by-one

Configuration	$4\pi\epsilon_0 W$
q_1	0
$+ q_2$	$+ q_1 q_2 / r_{12}$
$+ q_3$	$+ q_1 q_3 / r_{13} + q_2 q_3 / r_{23}$
$\dots$	$\dots$
$+ q_n$	$+ q_1 q_n / r_{1n} + q_2 q_n / r_{2n} + \dots + q_{n-1} q_n / r_{n-1,n}$

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- Potential energy of charge distribution ρ

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- We can consider the energy to be stored in the electric field with energy density

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Electrostatic energy for a collection of point charges

- Energy density from the field formula is positive definite

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- It just means ∞ energy is needed to pack finite charge into 0 volume (which we ignore in point charge formulas) using infinitesimal pieces

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- The potentials $\{V_i\}$ are linear in the charges

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Variational method

- Variational methods used for deriving equations of motion (eg. Hamilton's principle in mechanics)
- as well as computing approximate solutions
- Consider the *functional* I

$$I[\psi] = \int_V \left(\frac{1}{2} \vec{\nabla} \psi \cdot \vec{\nabla} \psi - g \psi \right) d^3x$$

- where $\psi(\vec{x})$ is well-behaved within volume V and on its boundary S and $g(\vec{x})$ is a *source* function with no singularities in V
- As we change ψ by $\delta\psi(\vec{x})$, I changes by

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$$\delta I = \oint_S \delta\psi \frac{\partial\psi}{\partial n} da - \int_V \left(\nabla^2\psi + g \right) \delta\psi d^3x$$

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$$\delta I = 0 \rightarrow \nabla^2\psi = -g \text{ in } V, \quad \delta\psi = 0 \text{ on } S$$

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- In order to solve an electrostatic problem with given bc, we start with a *trial function*

$$\psi(\vec{x}) = A \Psi(\vec{x}, \alpha, \beta, \dots)$$

with parameters A (for normalization), $\alpha, \beta, \dots$
that satisfies the bc on S

- and calculate

$$I[\psi] = I(A, \alpha, \beta, \dots)$$

- This transforms the problem into parameter fitting

$$\frac{\partial I}{\partial \alpha} = \frac{\partial I}{\partial \beta} = \dots = 0, \Rightarrow \{\hat{\alpha}, \hat{\beta}, \dots\}$$

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Example

Hollow circular cylinder (uniform in z) with BC $\psi(\rho = 1) = 0$ and source $g(\rho)$

$$\nabla^2 \psi = \frac{1}{\rho} \partial_\rho (\rho \partial_\rho \psi) = -g(\rho)$$

Polynomial trial function that satisfies $\psi(\rho = 1) = 0$

$$\begin{aligned}\psi_1 &= \alpha_1(1 - \rho) + \beta_1(1 - \rho)^2 + \gamma_1(1 - \rho)^3 \\ \frac{1}{2\pi} I_1(\alpha_1, \beta_1, \gamma_1) &= \frac{1}{4} \alpha_1^2 + \frac{1}{3} \alpha_1 \beta_1 + \frac{1}{4} \alpha_1 \gamma_1 + \frac{1}{6} \beta_1^2 \\ &\quad + \frac{3}{10} \beta_1 \gamma_1 + \frac{3}{20} \gamma_1^2 - (c_1 \alpha_1 + c_2 \beta_1 + c_3 \gamma_1) \\ c_n &= \int_0^1 g(\rho) (1 - \rho)^n \rho d\rho \\ \frac{\partial I_1}{\partial \alpha_1} &= \frac{\partial I_1}{\partial \beta_1} = \frac{\partial I_1}{\partial \gamma_1} = 0\end{aligned}$$

Example

Hollow circular cylinder (uniform in z) with BC $\psi(\rho = 1) = 0$ and source $g(\rho)$

$$\nabla^2 \psi = \frac{1}{\rho} \partial_\rho (\rho \partial_\rho \psi) = -g(\rho)$$

Polynomial trial function that satisfies $\psi(\rho = 1) = 0$

$$\begin{aligned} \psi_2 &= \alpha_2 \rho^2 + \beta_2 \rho^3 + \gamma_2 \rho^4 - (\alpha_2 + \beta_2 + \gamma_2) \\ \frac{1}{2\pi} I_2(\alpha_2, \beta_2, \gamma_2) &= \frac{1}{2} \alpha_2^2 + \frac{6}{5} \alpha_2 \beta_2 + \frac{4}{3} \alpha_2 \gamma_2 + \frac{3}{4} \beta_2^2 \\ &\quad + \frac{12}{7} \beta_2 \gamma_2 + \gamma_2^2 - (e_2 \alpha_2 + e_3 \beta_2 + e_4 \gamma_2) \end{aligned}$$

$$e_n = \int_0^1 g(\rho) (\rho^n - 1) \rho d\rho$$

$$\frac{\partial I_2}{\partial \alpha_2} = \frac{\partial I_2}{\partial \beta_2} = \frac{\partial I_2}{\partial \gamma_2} = 0$$

Example

For

$$g(\rho) = -5(1 - \rho) + 10^4 \rho^5 (1 - \rho)^5$$

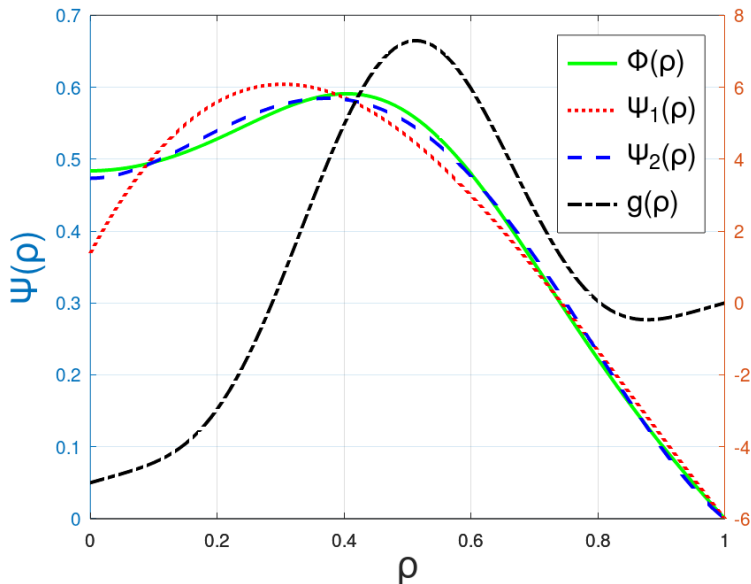
the parameters can be found as

$$\begin{array}{llll} c_1 = 0.4518 & \alpha_1 = 1.1092 & e_2 = -0.6654 & \alpha_2 = 2.9150 \\ c_2 = 0.1663 & \beta_1 = 0.5546 & e_3 = -0.8041 & \beta_2 = -7.0306 \\ c_3 = 0.0553 & \gamma_1 = -1.2944 & e_4 = -0.8813 & \gamma_2 = 3.6422 \\ & I_1 = -1.5136 & & I_2 = -1.5817 \end{array}$$

while the exact solution is

$$\begin{aligned} \Phi(\rho) &\simeq 69.444 \rho^{12} - 413.22 \rho^{11} + 1000 \rho^{10} \\ &\quad - 1234.6 \rho^9 + 781.25 \rho^8 - 204.08 \rho^7 \\ &\quad - 55.556 \rho^3 + 1.25 \rho^2 + 0.48379 \\ I[\Phi] &= -1.6017 \end{aligned}$$

Example



Harmonic functions

- Solutions $f(x_1, x_2, \dots, x_n)$ to Laplace eqn (in n dimensions)

$$\nabla^2 f = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \dots + \frac{\partial^2 f}{\partial x_n^2} = 0$$

- in an open subset of $\mathbb{R}^n$ are called *harmonic functions*
- f can not have a local maximum or minimum, the extrema are on the boundaries
- The mean of f inside or on the surface of any ball (hypersphere) is equal to its value at the center of the ball

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Relaxation method for 2D problems

Approximate soln of Poisson/Laplace eqn on a square grid using the mean value property – simple iterative procedure

- 1 Fix potential at points on the boundary (if Dirichlet bc)
- 2 Assign random initial values to other grid points
- 3 Replace the potential at each point (i, j) with the average $\langle\langle\Phi\rangle\rangle(i, j)$ of its neighbors

$$\langle\Phi\rangle_c(i, j) = \frac{1}{4} [\Phi(i-1, j) + \Phi(i+1, j) + \Phi(i, j-1) + \Phi(i, j+1)]$$

$$\begin{aligned}\langle\Phi\rangle_s(i, j) = \frac{1}{4} [\Phi(i-1, j-1) + \Phi(i-1, j+1) \\ + \Phi(i+1, j-1) + \Phi(i+1, j+1)]\end{aligned}$$

$$\langle\langle\Phi\rangle\rangle = \frac{4}{5}\langle\Phi\rangle_c(i, j) + \frac{1}{5}\langle\Phi\rangle_s(i, j)$$

- 4 Repeat until $\Phi(i, j)$ converges everywhere (change with each iteration below some limit)

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